

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Subject: 353152 Electronic Circuits I

Date: 2 August 2013

Year: 2013

Section: 5 - 6

Time: 13.00-15.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
 2. No documents are allowed to be taken out of the examination room.
 3. Text books and dictionaries are NOT allowed.
 4. Calculator is allowed.
 5. Electronic communication devices are NOT allowed during the examination.
 6. The examination has 11 pages (including this page), 2 parts (44 questions) and a total score of 100 points.
 7. Write all your answers on this exam sheet.
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Part 1 (80 points) Multiple choices

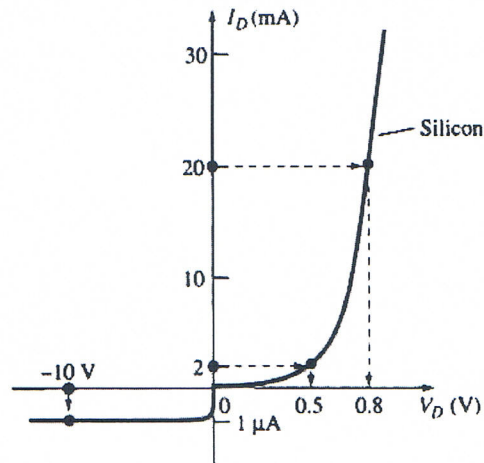
Instruction: Mark the correct answer for the following questions in the provided answer sheet.

1. Valence shell contain up to _____ electrons.
 (a) 2 (b) 4 (c) 6 (d) 8
2. In the atomic lattice the _____ and _____ form the nucleus.
 (a) electrons, neutrons (b) electrons, protons
 (c) neutrons, protons (d) None of the above
3. In n-type material the _____ is called the majority carrier.
 (a) electron (b) hole (c) proton (d) neutron
4. In what state is a silicon diode if the voltage drop across it is about 0.7 V?
 (a) No bias (b) Forward bias (c) Reverse bias (d) Zener region
5. The source voltage must be _____ the voltage drop across the diode to conduct the diode.
 (a) smaller (b) larger (c) the same as (d) zero
6. Which type of material has the biggest energy gap between conduction band and valence band?
 (a) conductor (b) semiconductor
 (c) insulator (d) superconductor
7. Atoms of semiconductors contain _____ valence electrons.
 (a) 2 (b) 4
 (c) 6 (d) 8
8. How much voltage drop over germanium diode when it is in forward bias mode?
 (a) 0.2 V (b) 0.3 V
 (c) 0.6 V (d) 0.7 V
9. Which one of following is true?
 (a) Diode is open circuit when operating in forward bias mode.
 (b) Forward bias means the voltage on p junction is lower than n junction.
 (c) For ideal diode, there is no voltage drop for forward bias mode.
 (d) No correct answer.

10. Calculate DC resistance R_{DC} of a diode having $I_D = 30 \text{ mA}$ and $V_D = 0.75 \text{ V}$.

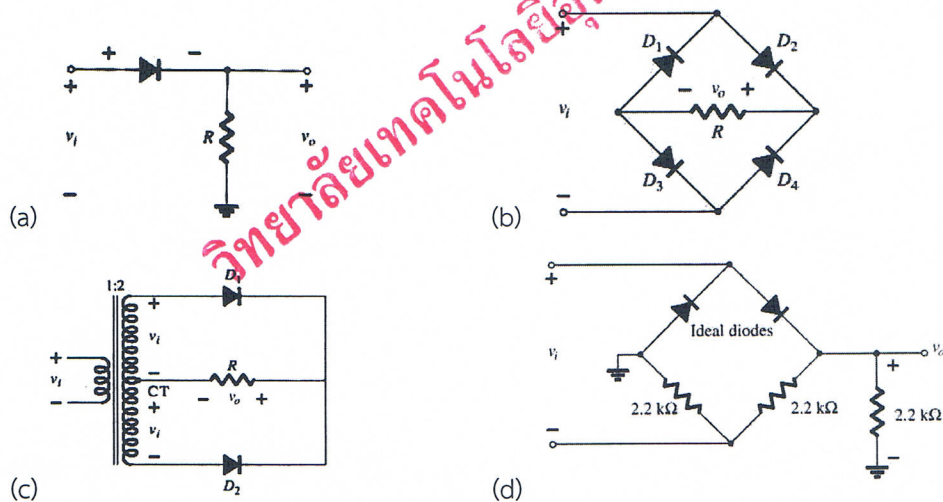
- (a) 25 (b) 40 (c) 0.04 (d) 0.025

11. From Figure below, what is the DC resistance at 20 mA?

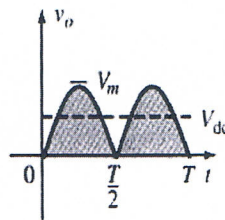


- (a) 4 m Ω (b) 4 Ω (c) 40 Ω (d) 40 k Ω

12. Which of the following is half-wave rectifier?

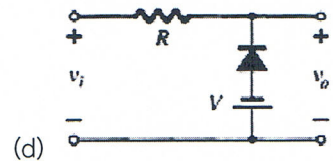
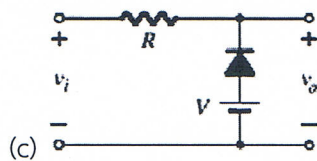
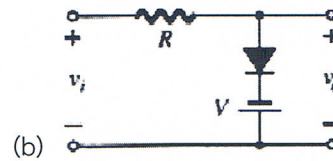
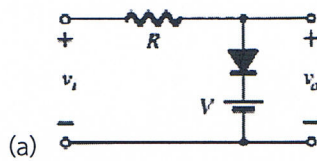
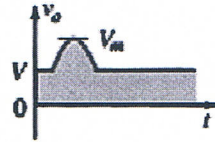


13. Which of the following is the average output value of full-wave rectifier?

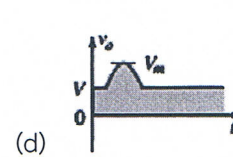
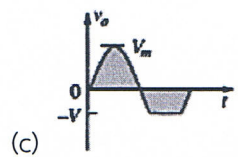
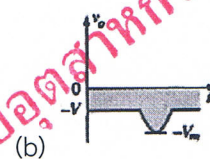
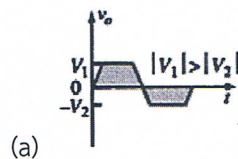
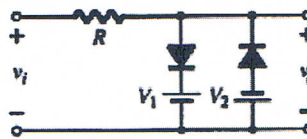


- (a) $0.318V_m$ (b) $0.5V_m$ (c) $0.636V_m$ (d) $1.414V_m$

14. Which circuit can generate output voltage as below?



15. What is the output voltage of the circuit below?



16. Which of following transistor operating mode represents the open circuit?

- (a) Active (b) Cut-off (c) Saturation (d) None

17. Which of following is true?

- (a) EB Junction is forward biased when transistor is in active mode.
 (b) EB Junction is forward biased when transistor is in cutoff mode.
 (c) CB Junction is reverse biased when transistor is in saturation mode.
 (d) CB Junction is forward biased when transistor is in cutoff mode.

18. The intersection of the load line with the characteristic curve determines the _____ of the system.

- (a) characteristic curve (b) forward bias
 (c) load-line analysis (d) point of operation

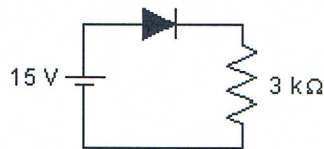
19. The load line is defined by the _____ and a characteristic curve is defined by the _____.

- (a) quiescent point, device (b) device, network dc network
(c) network, device (d) None of the above

20. A germanium diode is approximated by _____ equivalent for voltages less than 0.3 V.

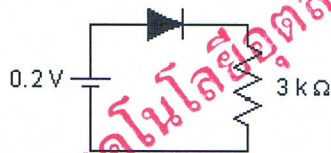
- (a) a short circuit (b) series circuit
(c) a parallel circuit (d) an open circuit

21. Determine the current level if $E = 15 \text{ V}$ and $R = 3 \text{ k}\Omega$, if the diode is assumed to be an ideal diode.



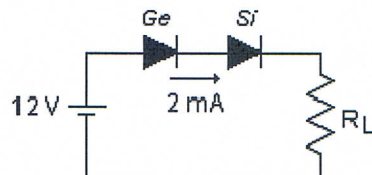
- (a) 0 A (b) 4.76 mA (c) 5 mA (d) 5 A

22. Determine the voltage across the resistor, if the diode is assumed to be an idea diode.



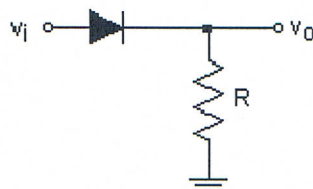
- (a) 0 V (b) 0.09 V (c) 0.2 V (d) 0.44 V

23. Determine the value of the load resistor.



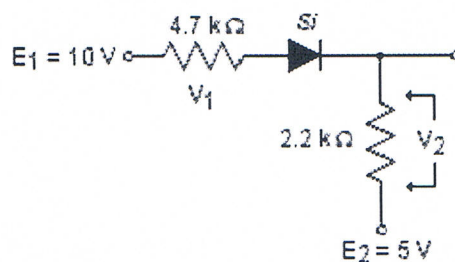
- (a) $R_L = 5 \text{ k}\Omega$ (b) $R_L = 5.5 \text{ k}\Omega$
(c) $R_L = 6 \text{ k}\Omega$ (d) None of the above

24. What best describes the circuit?



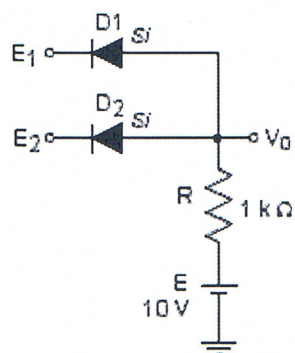
- (a) Full-wave rectifier (b) Bridge rectifier
(c) Clipper (d) Clamper

31. What is β_{dc} equal to?
 (a) I_B / I_E (b) I_C / I_E (c) I_C / I_B (d) I_B / I_C
32. Determine the value of α when $\beta = 100$.
 (a) 1.01 (b) 101 (c) 0.99 (d) 10.1
33. The cutoff region is defined by I_B ____ 0 A.
 (a) < (b) > (c) $\leq$ (d) $\geq$
34. Which of following transistor operating mode represents the open circuit?
 (a) Active (b) Cut-off
 (c) Saturation (d) None
35. At what region of operation is the base-emitter junction forward biased and the base-collector junction reverse biased?
 (a) Saturation (b) Active
 (c) Cutoff (d) None of all
36. Which of the following equipment can check the condition of a transistor?
 (a) Current tracer (b) Digital display meter (DDM)
 (c) Ohmmeter (VOM) (d) All of the above
37. The common-collector configuration has a ____ input impedance and a ____ output impedance.
 (a) low, high (b) high, low (c) high, high (d) low, low
38. Which of the following currents is nearly equal to each other?
 (a) I_B and I_C (b) I_E and I_C
 (c) I_B and I_E (d) I_B , I_C , and I_E
39. Determine V_2 .



- (a) 3.201 V (b) 0 V (c) 4.3 V (d) 1.371 V

40. Determine V_o if $E_1 = E_2 = 10\text{ V}$.



(a) 9.3 V

(b) 10 V

(c) -10 V

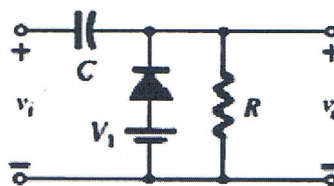
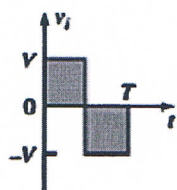
(d) 0 V

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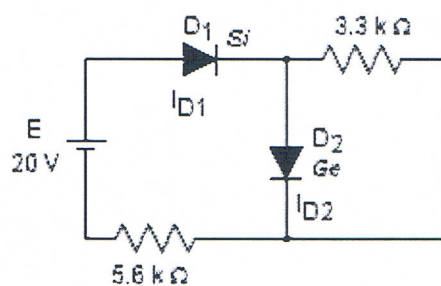
Part 2 (20 points) Calculation

Instruction: Show the mathematical expression and answers the following problems.

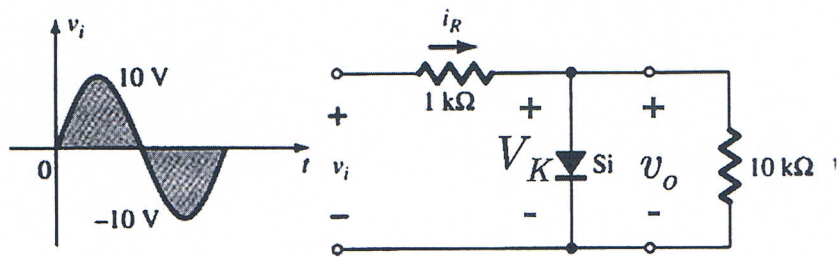
1. From following Figure, sketch one cycle of output.



2. Determine I_{D1} from figure below

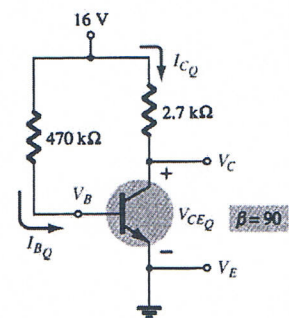


3. From Figure below, sketch one cycle of output.



4. From Figure below, determine I_B , I_C , V_{CE} , V_C , and V_B .

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Answer Sheet

Name: _____ ID: _____

Subject: 353152 Electronic Circuits I

	a	b	c	d
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	a	b	c	d
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